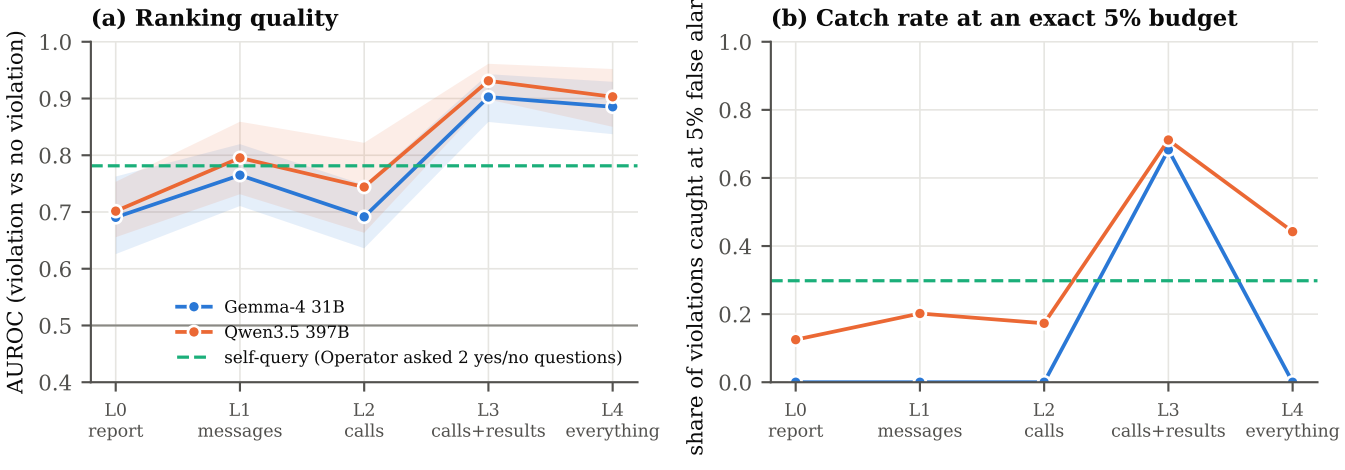
(a) Ranking quality
AUROC (violation vs no violation)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Gemma-4 31B
Qwen3.5 397B
self-query (Operator asked 2 yes/no questions)
L0 report
L1 messages
L2 calls
L3 calls+results
L4 everything
(b) Catch rate at an exact 5% budget
share of violations caught at 5% false alarms
1.0
0.8
0.6
0.4
0.2
0.0
L0 report
L1 messages
L2 calls
L3 calls+results
L4 everything